10-4 Data Analysis [3]

Point!

Variance and Standard Deviation

• Variance and standard deviation: These represent numerically how much the data is spread out from the mean ([1]**mean**). Let x_1, x_2, …, x_n represent the values of the data. $\bar{x}$ refers to the mean, and n refers to the number of data points.

(1) ([2]**Deviation**) : $x_1 - \bar{x}$, $x_2 - \bar{x}$, …, $x_n - \bar{x}$

The deviation is the value obtained by subtracting the mean from a data value.

(2) ([3]**Variance**) : The mean value of $(x - \bar{x})^2$. The ([4]**larger**) the variance, the more dispersed the data is.

In Google Sheets, the VARP function can be used to determine variance, while in Excel, the VAR.P function is used.

(3) ([5]**Standard deviation**) : $s = \sqrt{\text{Variance}}$

The standard deviation is the positive square root of the variance.

In spreadsheet software, the STDEV.P function can be used to determine the variance.

The standard deviation should be expressed in the same unit as the data.

<Steps for Calculating the Standard Deviation>

Summarize data in a table as shown on the right according to the following steps.

[1] Determine the ([6]**mean**) of the data.

[2] Find the ([7]**deviation**) for each individual data point.

[3] ([8]**Square the deviation**) of each individual data point.

[4] Determine the variance, which is the ([9]**mean**) of [3].

	x	$x - \bar{x}$	$(x - \bar{x})^2$
x_1		[2]	[3]
x_2		[2]	[3]
x_3		[2]	[3]
Total			
Mean	[1]		[4]

Variance

Warm Up

The following data represents the results of a handball throw by six individuals.

26, 25, 32, 28, 32, 25 (m)

(1) Determine the mean $\bar{x}$ of the data.

(2) Determine the variance and standard deviation of the data.

Explanation

(1) Summarize the data in a table like the one on the right.
The mean is 28 m.

(2) The variance is 9, and the standard deviation is $\sqrt{9} = 3$ (m).
3 m

	x	$x - \bar{x}$	$(x - \bar{x})^2$
x_1	26	-2	4
x_2	25	-3	9
x_3	32	4	16
x_4	28	0	0
x_5	32	4	16
x_6	25	-3	9
Total	168		54
Mean	28		9

Try

1 The following data represents the English vocabulary test scores of ten students.

9, 3, 4, 10, 10, 5, 7, 9, 10, 3 (points)

(1) Determine the mean of the data.

(2) Determine the variance and standard deviation of the data to two decimal places. However, let $\sqrt{2}$ = 1.414.

2 The following table shows the number of days it took five students to complete two types of assignments, A and B.

A	22	28	25	26	24
B	21	29	27	25	28

(1) Determine the variance for assignment A.

(2) Determine the standard deviation to two decimal places for assignment B. However, let $\sqrt{2}$ = 1.414.

(3) Answer which data set, assignment A or assignment B, has a greater variance.

Exercise

1 The following data represents the weights of eight harvested strawberries.

22, 25, 18, 17, 22, 21, 20, 15 (g)

(1) Determine the mean of the data.

(2) Determine the variance and standard deviation of the data.

2 The following table shows the test scores in math and English for five students.

Math	7	9	6	10	8
English	10	8	6	4	2

(1) Determine the variance of the math scores.

(2) Determine the standard deviation of the English test scores to two decimal places. However, let $\sqrt{2}$ = 1.414.

(3) Answer which data set has a greater variance, math or English.